

A scenic mountain landscape with green hills, rocky cliffs, and a cloudy sky. The foreground shows a rocky, grassy slope. In the middle ground, there are steep, rocky cliffs and green hills. The background features more distant, hazy mountain ranges under a sky filled with white and grey clouds.

AI & Software Engineering Workshop

Week 2, Day 5: Fine-tuning and Deployment

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The problem with a base model

Ask it: "Ո՞րն է Հայաստանի մայրաքաղաքը"

It might **continue** with more questions instead of answering, because continuing text is *all it knows*.

- **Base model** = autocomplete
- **Chatbot** = answers when asked

The bridge between them is **fine-tuning**.

Chat tokens: teaching it turns

Wrap every example in special tokens (*format sketch*):

```
<|user|>Ով է գրել «Գիքորը»:<|end|>  
<|assistant|>Հովհաննես Թումանյանը:<|end|>
```

The model learns the *pattern*: after `<|assistant|>` comes **an answer**, and `<|end|>` means stop.

Three new tokens, a whole new behavior.

Run it

```
python 1_download.py --download --qa          # Q&A sources
python 2_prepare.py --qa                      # merge + filter
python 3_tokenize.py --qa --tokenizer bpe
python 6_finetune.py --tokenizer bpe          # loads checkpoints/final.pt
```

- ~29K raw Q&A pairs → **~17K** after quality filtering
- A few hundred steps: **well under an hour**
- `chat_best.pt` keeps whichever step had the best validation loss

Talk to it

```
python 8_chat.py
```

- Ask it the capital question again: **now it answers**
- Try: a factual question, a creative request, something it can't know
- Be fair: it's 350M parameters, not Claude. Celebrate what works.

Ship it

```
python 7_deploy.py --repo <your-username>/armgpt
```

- Uploads weights + tokenizer + model card to HuggingFace
- The Gradio demo Space (like `edisimon/armgpt-demo`) serves it on CPU, a public URL anyone can try
- Your model now has a home on the internet

The full circle: into your Week 1 bot

The bot has a second-provider slot for any **OpenAI-compatible** endpoint:

```
ARMGPT_BASE_URL=https://<your-endpoint>/v1  
ARMGPT_API_KEY=<key>  
ARMGPT_MODEL=armgpt
```

Restart → a `/model` command appears → `/model armgpt` ⇔ `/model main`

- The provider sends **only your latest message** (no system prompt, no memory), sized for a small self-trained model
- *(The Space is the demo UI; the bot needs the API endpoint, reference setup runs on Modal)*

Demo day: run it locally

```
make run          # in your Week 1 bot repo
```

- PythonAnywhere's free tier can't reach the model endpoint (outbound whitelist), so today the bots run locally on your computers
- Try everyone's bot: `/model armgpt`, same question to both brains
- One question, two answers: a 350M model **you trained**, and a frontier model, side by side in *your* bot

Two weeks, full stack

Week 1: you shipped a bot to the internet.

Week 2: you built the intelligence inside one.

Code → Train → Deploy. You've now done all three.

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